

MT Joint Pre-Colloquium Session 1

Exploring the landscape of data science
thesis topics



Welcome!

Welcome!

Why you are here today

- Get to know your potential supervisors better
- Get inspired for your master's thesis proposal
- Learn about opportunities to work on topics, research questions, methods, and data within the domain of data science faculty

Why we are here today

- Get an informal conversation about your thesis projects started
- Pitch some ideas that we think would be great to work on

The Master's Thesis Proposal

1. What is your research topic?

A comprehensive list of academic fields relating to the governance and policy domain is provided to choose from. You may choose up to three domains which most reflect your interest to develop a thesis project.

2. What is your research question?

At this stage your research question will not be fully formed, and you may currently be considering more than one question. Your question will also change and evolve throughout the project. Nevertheless, it is important to form the question as early as possible. Please list your key question(s) of interest.

3. Why would you like to explore this question?

Please let us know why you are curious about your research topic, and why you think your question(s) are worth exploring.

4. What are the policy questions to which your thesis will relate?

You should clarify the relevance of your thesis by relating it to the literature. You may want to cite theoretical, empirical, normative, and policy-related debates.

5. What is your proposed methodology?

Provide detail about the methodology/ies you will employ to answer your research questions.

6. What type of data do you need?

A main corner stone of your thesis is having high-quality data to work with. Ensure that the data you need are available and theoretically suitable to answer your research question(s). **List those data sources in your proposal.**

7. Who are your preferred supervisors?

(You must indicate a 1st, 2nd and 3rd choice)

Envisaged timeline

Colloquium dates	Format	Session content
Session 1: Mid-September 2024	Joint session (optional; not part of official colloquium)	Exploring the landscape of data science thesis topics and ideas
Session 2: Mid-September 2024	Joint session (optional; not part of official colloquium)	From thesis idea to proposal: the art of developing a research question
Master's thesis proposal supervision – Deadline October 1, 2024, via Moodle		
Session 3: Mid-October 2024	Supervisor session (mandatory)	Presentation and discussion of thesis proposals
Session 4: Mid-November 2024	Joint session (mandatory)	Developing the structure and research design of your thesis
Session 5: Early February 2025	Joint session (mandatory)	Notes on scientific writing for publication
Session 6: Early February 2025	Supervisor session (mandatory)	1:1 meetings
Session 7: Mid-March 2025	Supervisor session (mandatory)	1:1 meetings

Potential supervisors

Core data science team



Prof. Drew Dimmery, PhD
Professor of Data Science for the
Common Good



Prof. Lynn Kaack, PhD
Assistant Professor of Computer
Science and Public Policy



Dr. William Lowe
Senior Research Scientist



Prof. Asya Magazinnik, PhD
Professor of Social Data Science



Prof. Dr. Simon Munzert
Professor of Data Science and Public
Policy | Director, Data Science Lab

Faculty with quantitative research profile, including



Prof. Arianna Ornaghi, PhD
Assistant Professor of Economics



Prof. Dr. Christian Traxler
Professor of Economics



Prof. Dr. Mujaheed Shaikh
Professor of Health Governance



Prof. Egon Tripodi, PhD
Assistant Professor of Economics



Prof. Dr. Anita Gohdes
Professor of International and Cyber
Security



**Prof. Dr. Julian
Wucherpfennig**
Professor of International Affairs and
Security



Prof. Ruth Ditzmann, PhD
Professor of Psychology and Public
Policy (currently on maternity leave)



Prof. Mark Kayser, PhD
Professor of Applied Methods and
Comparative Politics

Academic Counsellor

Our Academic Counsellor, **Maria Uttenthal**, is here to help you with anything academic. Overall, the counselling sessions centre around 3 topics:

1. **Master's thesis support:** Basically all parts of the research process - brainstorming topic ideas, choosing the right methodology, data collection, tips and tricks for efficient writing, etc. These sessions complements the master's thesis workshops that Maria runs in parallel.
2. **Learning and study support:** Struggles with time management, meeting deadlines, prioritising and managing your workload.
3. **Designing your curriculum:** Choosing the area of concentration, elective courses, skills and trainings.

Watch out for the **thesis info** sessions Maria offers!

1:1 sessions with her can be booked via Moodle. That's also where you find plenty more materials to boost your thesis writing and methods skills.



**Drew's topic
suggestions**

Drew's profile

Methods and data

- Causal inference, impact evaluation
- Experiments (online, survey, field)
- Statistics
- Measurement
- Web and social media data
- Machine learning
- *Causal* Machine Learning

Applied Research themes

- The Internet
- Social Media
- AI & Society

Other Current Projects

- On parental leave
- Will provide active feedback, but expect more of it asynchronously (email, Teams, etc)



Methods focused projects

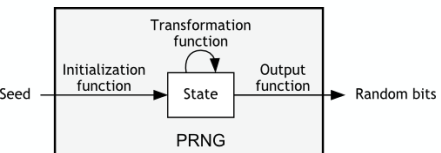
- **Make an existing experimental design randomized**
 - The design takes a MST over the data and then emits the optimal design on the MST
 - Add randomization into this: sampling spanning trees, random perturbations of the input data

- **Smarter sampling of points for minibatches in SGD**

- For example, leverage score sampling or more complex experimental design methods.
- Can be easily combined with a project focused on supervised learning

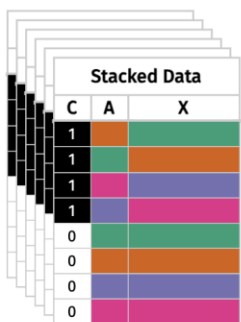
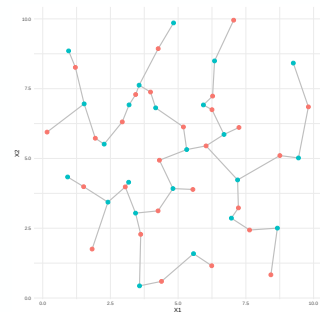
- **Can you learn a pseudo-random number generator with a neural network?**

- Just try to predict the next value from a PRNG
- Could also think about doing things like taking a regression dataset, ordering it according to its label and then randomly selecting units based on the PRNG, including row numbers.
- This has strong implications for training large language models (is there leakage via the PRNG?).



- **Develop a “permutation weighting” model using neural networks and SGD**

- A flexible CausalML method to convert weighting problems into supervised learning problems
- Using SGD makes it much more scalable for large datasets
- Write software to do this, evaluate how well it does, see how tuning parameters matter



Applied ML/CausalML projects

- A good Applied ML project will have a good answer to a “Why?” question.
 - Why does someone need a prediction of the label?
 - What problem does it help solve?
- Measure conversation quality using “Graph Motifs”
 - Look at sequences of replies to see if users are engaging positively with each other
 - Would be a great opportunity to build complex neural networks
- Compare modelling strategies on the US2020 Facebook and Instagram Election Project
 - RQs:
 - Can we detect subgroups for which there are larger effects?
 - Can we measure heterogeneous effects more precisely?
 - Is it possible to use better models and find such heterogeneity?
 - I’m in the process of requesting data, so there is some uncertainty on when it will be available



One Million Posts Corpus



**Simon's topic
suggestions**

Simon's profile

Methods and data

- Experiments (online, survey, field)
- Survey design
- Measurement
- Causal inference, impact evaluation
- Multilevel modelling
- Meta-analysis
- Survey data
- Behavioral (web) data
- Web and social media data

Research areas

- Political behavior and voting
- Public opinion
- Online political communication

Sample of previously supervised thesis topics

- Examining the effects of female political representation in Brazilian municipalities (massive administrative data, RDD)
- Indian legislators and the Wikipedia marketplace of information (Wikipedia/Wikidata metadata, time-series analysis)
- Do Natural Disasters Affect Electoral Outcomes? (meta-analysis)
- Between deliberation and Polarization: Analyzing patterns of Germans' online media diets (trace data)
- Privacy and Democracy in a Cambridge Analytica World: Predicting Party Choice from Browsing History (trace data, ML)
- The distant impact of nuclear disasters: Consequences of the Fukushima accident on German political preferences
- YouTube News and Politics Media Diets: A User-based Analysis of Filter Bubbles and the Recommendation Algorithm
- An Analysis of Hate and Offensive Speech against Greta Thunberg on Twitter
- Leveraging survey and web-tracking data for audience networks (trace data, network analysis)
- Predicting voting behavior from search engine queries (trace data, ML)

Simon's suggested topics

Work with data I've collected

- Online media use and preferences for hate speech regulation
- Global attention towards climate change and public health

Synthesize existing research

- Meta-analysis of the effects of social media on anti-social behavior
- Cheating in surveys: A meta-analysis
- Meta-analysis of COVID-19 contact tracing app usage

Collect your own (survey) data

- Scientific literacy and trust in science
- What do policymakers know about data science and AI?

Meta-analysis

- Quantitatively synthesize findings
- Goal is to statistically pool estimates across studies; potentially also look into study-level factors that might drive results
- Benefits:
 - You get a systematic overview of a field
 - You piggyback on other people's research to produce relevant results
- Challenges:
 - Requires learning about meta-analysis and its pitfalls

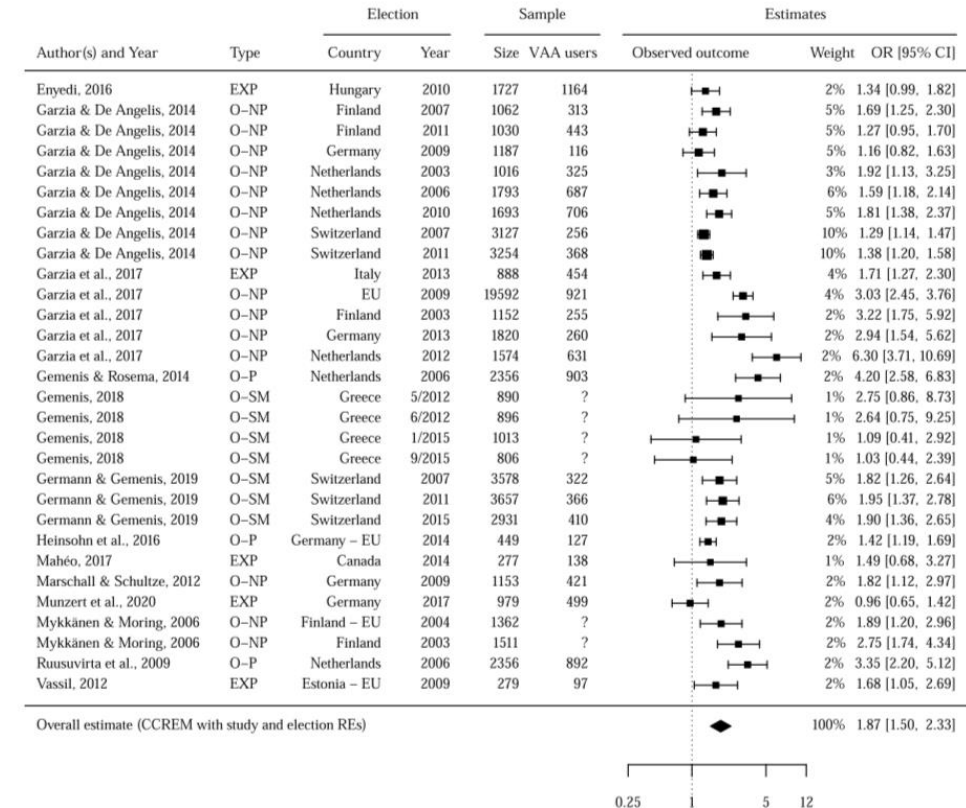


Figure 3. Effect sizes (odds ratios) of VAA usage on turnout.

Collaborative survey project

- Co-design a survey to collect your own data on public preferences, attitudes, knowledge, behaviors
- Survey experiments (e.g., framing, exposure, conjoint) encouraged
- Benefits:
 - You get to collect your own data!
- Challenges:
 - Requires lots of coordination with others; additional data collection effort
 - Requires learning about the do's and don'ts of survey design and analysis



Table 1 Patient attributes.

Attribute	Randomized categories
Gender	Male, Female
Age	30 years old, 44 years old, 64 years old
Chance of survival in ICU	20%, 50%, 80%
Occupation	Medical doctor, professor, nurse, cook
Children	Has school-aged children, has no children
Migration status	German citizen, migrant with residence permit
Criminal record	Has criminal record, has no criminal record

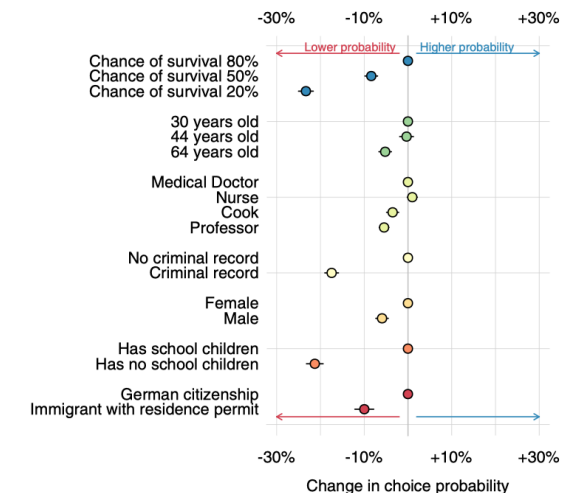


Fig. 1 For each patient attribute, the reference category is listed first and represented at 0 on the x-axis (80% chance of survival, 30 years old, Medical Doctor, No criminal record, Female, Has school-aged children, German citizen). For the other categories, dots (with 95% confidence intervals) are the difference in the average marginal component effects (AMCE) between the reference category and the category listed on the y-axis. Pooled survey waves 1–23.

Other potential areas from my team



Sascha Göbel
Public preferences,
Measurement,
Political elites



Sebastian Ramirez Ruiz
Evidence consumption,
Social media,
Political elites



Elias Koch
Parties and polls,
Election forecasting,
Experiments

**Lynn's topic
suggestions**

Lynn's profile

AI and Climate Technology Policy Group

Methods and data

- Statistics (with focus on prediction)
- Machine learning (generally)
- NLP
- Computer vision, remote sensing
- Forecasting
- Policy analysis
- Listed a number of transportation- and energy-related datasets in supervision plan

Research areas

- Climate policy in energy-related sectors, technology policy
- Transportation, buildings, electricity sector, climate + health
- Intersection of AI and ML with climate change
- The greenhouse gas emissions impacts of machine learning
- Municipal governance of AI for sustainability
- Machine learning methods for public policy

Example thesis: What resolution should satellite imagery have to detect electricity infrastructure?

- Partner: University of Edinburgh
- Data on the locations of electricity infrastructure are often not available, particularly in low-income countries.
- Machine learning has proven itself as a reliable method to infer energy infrastructure from satellite imagery.
- **RQ:** What resolution does satellite imagery need to have to detect electricity infrastructure?

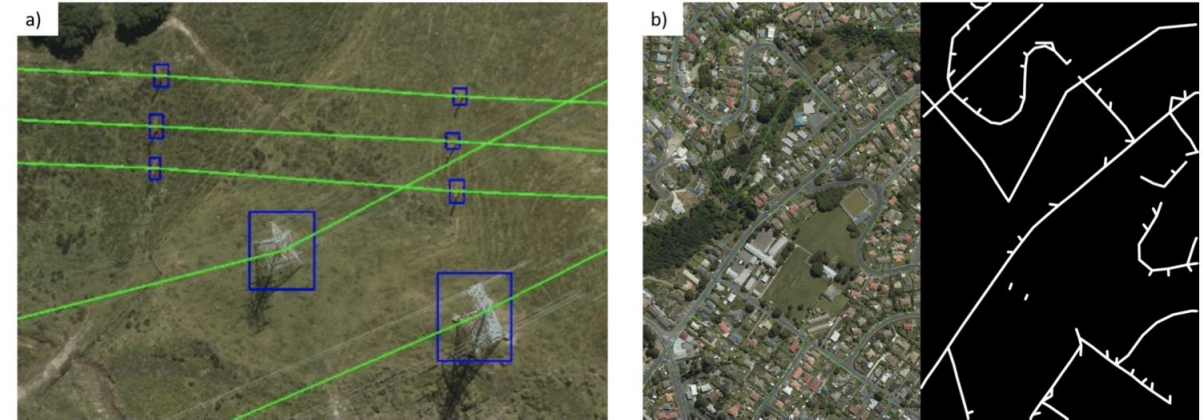


Fig. 3. Illustration of the PG inference task, where (a) shows the annotated tower and lines and (b) demonstrates the power grid in larger scale. Note the complexity of the topology in (b).

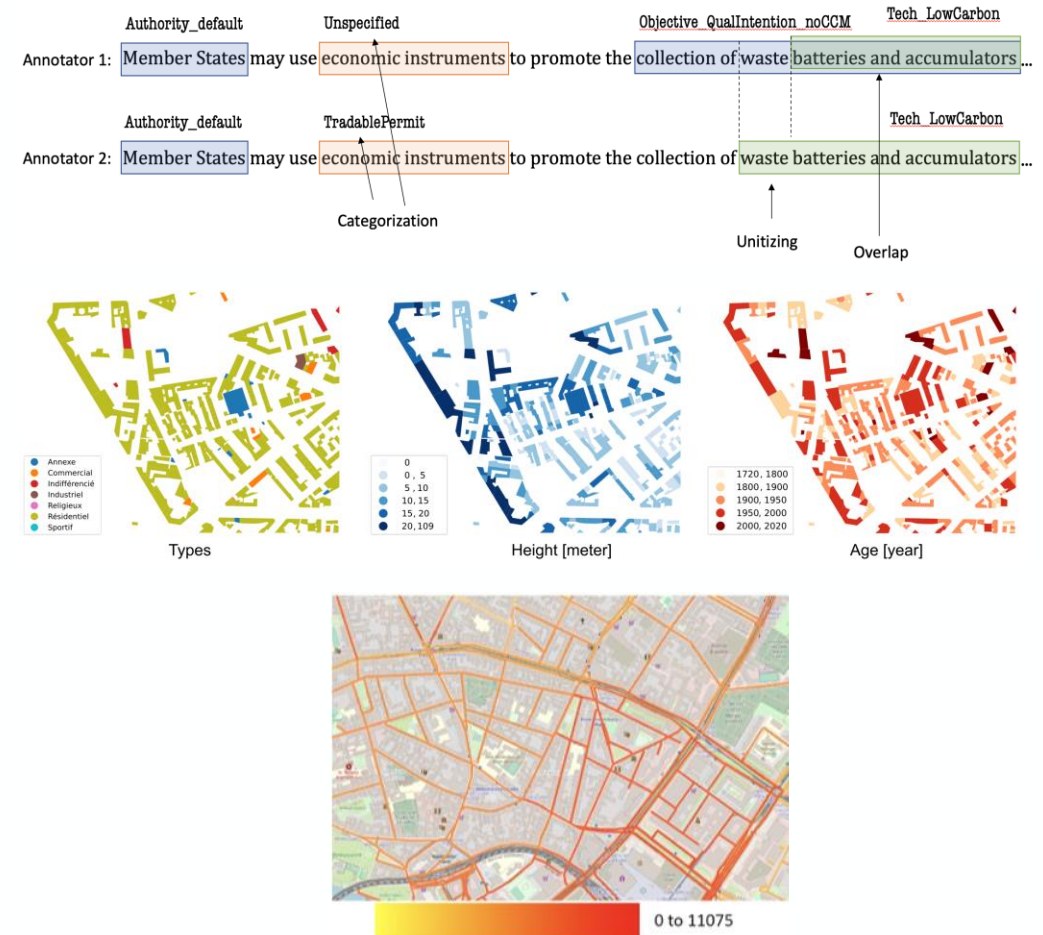
Lynn's suggested topics

See supervision plan for details

- Forecasting Cycling Volume Measurements: A Comparative Analysis of Various Techniques (with Silke Kaiser)
- Using machine learning for to model power plant operation (with Chiara Fusar Bassini)
- Where and how are big tech companies collaborating with cities on projects using machine learning for climate action? (with Josefine Hintz)
- Bridging the skill gap in local governments: What forms of capacity building exist for municipalities to acquire AI expertise? (with Josefine Hintz)
- In which ways can the use of AI increase greenhouse gas emissions?
- NLP on POLIANNA dataset for evaluating policy design
- Where do we need technology-specific regulatory approaches for minimizing greenhouse gas emissions associated with AI or digitalization?

Datasets in AI and Climate Technology Policy Group

- POLIANNA dataset (EU Laws annotated for policy design): <https://www.nature.com/articles/s41597-023-02801-z>
- EUBUCCO dataset (European building stock characteristics): <https://www.nature.com/articles/s41597-023-02040-2>
- Bike-sharing data Berlin from Nextbike and Call-a-Bike for 2019 and 2022 <https://zenodo.org/records/10046531>
- Power plant generation and (un)availability for Germany and Europe (entso-e and EEX)



Other potential areas from my team



Silke Kaiser

- Bike data
- Machine learning
- Statistics & econometrics
- Economics



Josefine Hintz

- Urban AI governance
- Structured interviews
- Systematic maps/reviews
- Digital governance



Chiara Fusar Bassini

- Power plant data
- Machine learning
- Optimization
- Electricity industry



Carol Sobral

- Mode share data
- Data science
- Fiscal policy

**Will's topic
suggestions**

Will's profile

Methods and data

- Statistics! Classical and Bayesian
- Research design and causal inference
- Measurement, particularly 'ideal point' estimation
- Text analysis: statistical models, and mixed methods

Research areas (also methods...)

- Spatial models of ideology / policy preferences from observational data: limitations and extensions
- Directed acyclic graphs: limitations and extensions
- The machine learning / causal inference interface
- Causal theories of fairness and bias, particularly in ML

Will's suggested topics

Previous supervision topics

- Legal text classification for competition impact assessment
- Bias mitigation in static word embeddings
- Sensor data for early wildfire detection
- Scaling speeches and tweets from German legislators
- Earthquake damage assessment using deep learning on satellite imagery
- Information and negativity in campaign advertising
- War crimes and corporate responsibility

Methodological research projects

- Causal inference in systems with feedback
- Text scaling with ideal points on covariates
- Dimensionality and other geometric issues in ideological spaces
- Bias and bias removal in word embeddings
- Causal and counterfactual theories of bias and fairness
- Causal critiques of multiverse and 'many labs' analyses
- Explainable AI

Other projects may be developed together in my team (e.g., on bike infrastructure, urban sustainability, mining...) ²⁷

Asya's topic suggestions

Asya's profile

Research Interests

- Anything to do with borders (national or subnational) and how they affect people's lives
- The drawing of political boundaries, such as voting districts, and how that relates to democratic representation
- How people are distributed over space, including by race and ethnicity, poverty, language, political ideology... (and why it matters)
- Spatial inequality: structural geographic (dis)advantage in resources, social service and public goods provision
- Innovations in conjoint analysis and other preference elicitation techniques
- Federalism (American and European)

Policy Areas

- Housing, immigration, policing, voting rights, education policy

Data and Methods

I can provide >100 maps of U.S. city council electoral districts for redistricting simulation work

- I am happy to provide input into survey design, or to collaborate on fielding surveys about political preferences
- Panel data, administrative data
- Projects combining game theory with empirical work

Successful projects previously advised

Asking Causal Questions

- An impact evaluation of the 2016 Colombian Peace Agreement with the FARC on deforestation
- Does a state's susceptibility to climate change play a role in the duration of its conflicts, and how do migration patterns and colonial history help explain this relationship?
- Disentangling the direction of causality between support for right-wing populist parties and climate change concern

Applying a Statistical Methodology to New Data

- Using Multiple Systems Estimation to get more reliable estimates of migrant fatalities crossing the US-Mexico border
- Using an automated redistricting algorithm to imagine what German electoral maps would look like if foreign residents could vote

Outlook

Outlook

Vetting your topic and research question

- (Why) is this relevant?
- Does the scope of the question seem adequate?
- What data is required to tackle the problem, and where do you get it?
- What are likely key challenges for data collection, measurement, modelling?

Up next

- MT Joint Colloquium Session 2: From thesis idea to proposal: the art of developing a research question
- Sep 20, 2024, 2-3.30pm, Room TBA